

Cheng Zhang

✉ zhangchengge@gmail.com | 🌐 holmes969.github.io/

Education

University of California, Irvine

PH.D. IN COMPUTER SCIENCE

Supervisor: Prof. Shuang Zhao

Irvine, CA

September 2017 – July 2022

Columbia University in the City of New York

M.S. IN COMPUTER SCIENCE

GPA: 3.97/4.0

New York, NY

September 2015 – February 2017

Beijing University of Technology

B.E. IN ELECTRONICS INFORMATION ENGINEERING

Beijing, China

September 2011 – June 2015

- GPA: 3.74/4.0

- Exchange student at **University of Illinois at Chicago** (senior year)

Publications

LSRM: HIGH-FIDELITY OBJECT-CENTRIC RECONSTRUCTION VIA SCALED CONTEXT WINDOWS

Zhengqin Li, **Cheng Zhang**, Jakob Engel, Zhao Dong

European Conference on Computer Vision (ECCV), 2026

ART: ARTICULATED RECONSTRUCTION TRANSFORMER

Zizhang Li, **Cheng Zhang**, Zhengqin Li, Henry Howard-Jenkins, Zhaoyang Lv, Chen Geng, Jiajun Wu, Richard Newcombe, Jakob Engel, Zhao Dong

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

IMAGE-SPACE ADAPTIVE SAMPLING FOR FAST INVERSE RENDERING

Kai Yan, **Cheng Zhang**, Sébastien Speierer, Guangyan Cai, Yufeng Zhu, Zhao Dong, Shuang Zhao

ACM Transactions on Graphics (SIGGRAPH), 2025

LIRM: LARGE INVERSE RENDERING MODEL FOR PROGRESSIVE RECONSTRUCTION OF SHAPE, MATERIALS AND VIEW-DEPENDENT RADIANCE FIELDS

Zhengqin Li, Dilin Wang, Ka Chen, Zhaoyang Lv, Thu Nguyen-Phuoc, Milim Lee, Jia-Bin Huang, Lei Xiao, **Cheng Zhang**, Yufeng Zhu, Carl S Marshall, Yuheng Ren, Richard Newcombe, Zhao Dong

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025

DIGITAL TWIN CATALOG: A LARGE-SCALE PHOTOREALISTIC 3D OBJECT DIGITAL TWIN DATASET

Zhao Dong, Ka Chen, Zhaoyang Lv, Hong-Xing Yu, Yunzhi Zhang, **Cheng Zhang**, Yufeng Zhu, Stephen Tian, Zhengqin Li, Geordie Moffatt, Sean Christofferson, James Fort, Xiqing Pan, Mingfei Yan, Jiajun Wu, Carl Yuheng Ren, Richard Newcombe

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025

ESTIMATING UNCERTAINTY IN APPEARANCE ACQUISITION

Zhiqian Zhou, **Cheng Zhang**, Zhao Dong, Carl Marshall, Shuang Zhao

Eurographics Symposium on Rendering (EGSR), 2024

TEXTUREDREAMER: IMAGE-GUIDED TEXTURE SYNTHESIS THROUGH GEOMETRY-AWARE DIFFUSION

Yu-Ying Yeh, Jia-Bin Huang, Changil Kim, Lei Xiao, Thu Nguyen-Phuoc, Numair Khan, **Cheng Zhang**, Manmohan Chandraker, Carl S Marshall, Zhao Dong, Zhengqin Li

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024

NEURAL-PBIR RECONSTRUCTION OF SHAPE, MATERIAL, AND ILLUMINATION

Cheng Sun, Guangyan Cai, Zhengqin Li, Kai Yan, **Cheng Zhang**, Carl Marshall, Jia-Bin Huang, Shuang Zhao, Zhao Dong

International Conference on Computer Vision (ICCV), 2023

SAM-RL: SENSING-AWARE MODEL-BASED REINFORCEMENT LEARNING VIA DIFFERENTIABLE PHYSICS-BASED SIMULATION AND RENDERING

Jun Lv, Yunhai Feng, **Cheng Zhang**, Shuang Zhao, Lin Shao, Cewu Lu

Robotics: Science and Systems (Best System Paper Award Finalist), 2023

EFFICIENT PATH-SPACE DIFFERENTIABLE VOLUME RENDERING WITH RESPECT TO SHAPES

Zihan Yu, **Cheng Zhang**, Olivier Maury, Christophe Hery, Zhao Dong, Shuang Zhao

Computer Graphics Forum (Eurographics Symposium on Rendering), 2023

EFFICIENT DIFFERENTIATION OF PIXEL RECONSTRUCTION FILTERS FOR PATH-SPACE DIFFERENTIABLE RENDERING

Zihan Yu, **Cheng Zhang**, Derek Nowrouzezahrai, Zhao Dong, Shuang Zhao

ACM Transactions on Graphics (*SIGGRAPH Asia 2022*), 41(6)

ANTITHETIC SAMPLING FOR MONTE CARLO DIFFERENTIABLE RENDERING

Cheng Zhang, Zhao Dong, Michael Doggett, Shuang Zhao

ACM Transactions on Graphics (*SIGGRAPH 2021*), 40(4)

PATH-SPACE DIFFERENTIABLE RENDERING OF PARTICIPATING MEDIA

Cheng Zhang^{*}, Zihan Yu^{*}, Shuang Zhao (^{*}equal contribution)

ACM Transactions on Graphics (*SIGGRAPH 2021*), 40(4)

PATH-SPACE DIFFERENTIABLE RENDERING

Cheng Zhang, Bailey Miller, Kai Yan, Ioannis Gkioulekas, Shuang Zhao

ACM Transactions on Graphics (*SIGGRAPH 2020*), 39(4)

MULTI-SCALE APPEARANCE MODELING OF GRANULAR MATERIALS WITH CONTINUOUSLY VARYING GRAIN PROPERTIES

Cheng Zhang, Shuang Zhao

Eurographics Symposium on Rendering (*EGSR*), 2020

A DIFFERENTIAL THEORY OF RADIATIVE TRANSFER

Cheng Zhang, Lifan Wu, Changxi Zheng, Ioannis Gkioulekas, Ravi Ramamoorthi, Shuang Zhao

ACM Transactions on Graphics (*SIGGRAPH Asia 2019*), 38(6)

FontCode: Embedding Information in Text Documents Using Glyph Perturbation

Chang Xiao, **Cheng Zhang**, Changxi Zheng

ACM Transactions on Graphics (*SIGGRAPH 2018*), 37(2)

Patents

SYSTEMS AND METHODS FOR STEGANOGRAPHY BASED ON TEXT FONTS

Changxi Zheng, Chang Xiao, **Cheng Zhang**

US Patent 10,755,375

METHOD AND DEVICE FOR EFFICIENT BUILDING FOOTPRINT DETERMINATION

Lincan Zou, Liu Ren, Zeng Dai, **Cheng Zhang**

US Patent 10,553,025

Employment

Meta Reality Labs

RESEARCH SCIENTIST

Topic: Computer graphics research (e.g., differentiable rendering, inverse rendering), 3D reconstruction

Redmond, WA

July 2022 - Present

NVIDIA Corporation

APPLIED DEEP LEARNING RESEARCH (ADLR) GROUP: RESEARCH INTERN

Topic: Differentiable rendering on GPU

Remote

June 2021 - November 2021

Facebook Reality Labs

GRAPHICS GROUP: RESEARCH INTERN

Topic: 3D reconstruction, differentiable rendering

Redmond, WA

June 2019 - September 2019

Robert Bosch LLC

HMI GROUP: MOTION CAPTURE AND RECOGNITION INTERN

Topic: Automatic cycle detection and similarity evaluation of time series data

Sunnyvale, CA

June 2018 - September 2018

Robert Bosch LLC

HMI GROUP: GRAPHICS RESEARCH INTERN

Topic: Building footprints determination for real-time shadow generation

Palo Alto, CA

May 2017 - August 2017

Columbia University

COMPUTER GRAPHICS GROUP: GRADUATE RESEARCH ASSISTANT

Topic: Text editing, information embedding

New York, NY

June 2016 - April 2017

Reviewer

2026	ACM SIGGRAPH, Computers & Graphics, ACM SIGGRAPH Asia (Technical Papers Committee)
2025	ACM SIGGRAPH, ACM SIGGRAPH Asia
2024	ACM SIGGRAPH, ACM SIGGRAPH Asia, EGSR, Journal of Imaging
2023	Nature Scientific Reports, Computers & Graphics, ACM SIGGRAPH Asia, Pacific Graphics
2022	ACM SIGGRAPH
2021	ACM SIGGRAPH, Computer Graphics Forum
2020	ACM SIGGRAPH Asia

Honors & Awards

2023	ACM SIGGRAPH Outstanding Doctoral Dissertation Award	Worldwide
2021	Facebook Fellowship	Worldwide
2013	Beichuan Academic Scholarship	Beijing University of Technology
2011-2014	Academic Excellence Award	Beijing University of Technology

Skills

Programming	C, C++, Python, Matlab, T _E X, CUDA
Software and Applications	Mitsuba, PyTorch, Unity3D, Blender, OptiX